

Jimmy Han

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Education

Rutgers University - New Brunswick, NJ

Jan. 2025

Bachelor's in Computer Science (*School of Arts and Sciences*)

- Relevant Coursework: Machine Learning Principles, Deep Learning, Principles of Info and Data Management, Intro to Artificial Intelligence, Data Structures, Discrete Structures, Design and Analysis of Computer Algorithms, Software Methodology, Linear Algebra

Experience

Database Developer Intern - Bergen School of Business

July - Aug. 2024

- Designed, developed, and managed a relational database in SQL to store and manage student and realtor information
- Leveraged SQL join statements, grouping, aggregations, and partitions to meticulously navigate through the database
- Integrated the database with a full stack Java application, supporting functionality for data organization, searching and sorting individuals by keywords, and performing aggregations. This streamlined the data storage and retrieval process, which reduced reliance on manual Excel workflows
- Ensured data consistency and optimized query performance to support seamless information access and reporting

Certifications

Microsoft Certified: Power BI Data Analyst Associate (PL-300)

Feb. 2025

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

June 2025

Databricks Certified Data Analyst Associate

June 2025

Oracle Database SQL Certified Associate (1Z0-071)

Apr. 2025

Tableau Desktop Specialist

Mar. 2025

Technical Skills

SQL, Python, PySpark, Spark, NumPy, Pandas, Pytorch, RegEx, Matplotlib, Scikit-Learn, Java, HTML/CSS

Power BI, Tableau, Excel, Fabric, Azure Databricks, Azure Data Factory, Microsoft SQL Server, VS Code, IntelliJ, Eclipse, Jupyter Notebook

Conceptual: ETL/ELT Pipelines and Orchestration, Dataflows, Data Warehousing, Data Visualization, Power Query

Projects

Power BI Report/Dashboard Portfolio | Power BI, DAX, Power Query

Mar. - May 2025

- Built a comprehensive **Retail Inventory Performance Report** to monitor revenue and units sold across time, multiple regions, categories, and store locations using interactive filters, KPIs, and dynamic visuals.
- Designed inventory health analytics using stock-to-sales ratios, forecast accuracy, and inventory vs. sales trend lines.
- Developed many DAX measures and calculated columns for time-based aggregation and granular segmentation.
- Transformed and cleansed multi-source data in Power Query for consistent schema and data-type corrections.
- View at github.com/Jimmy1634/PowerBI-Retail-Inventory-Report and github.com/Jimmy1634/Shoe-Retail-Dashboard

Azure Data Engineering Project | Power BI, Azure Databricks, PySpark | github.com/Jimmy1634/Flight-Delay-Analysis

June 2025

- Constructed an ETL pipeline to process and analyze flight delay data using Azure Storage and Databricks, depositing results into Data Lake tables. Processing followed medallion architecture using PySpark and SQL functions.
- Designed bronze, silver, and gold Spark dataframes, which were implemented as Delta tables for visualization.
- Created a dashboard in Power BI to visualize insights such as top performing airlines, most delayed locations, and average flight delays by airline. Used Power Query and custom measures/DAX expressions to support visuals.

LeNet5 Neural Network | Python - PyTorch, Scikit-Learn, Pandas, Matplotlib | github.com/Jimmy1634/LeNet_MNIST

Nov. 2024

- Manually implemented the LeNet-5 convolutional neural network from scratch to classify images of handwritten digits (0-9)
- Improved **model accuracy** from **75.16% to 92.04%**, a **+16.88% gain**, through applied data augmentation, input normalization, and architectural changes (ReLU activation, max pooling/dropout layers, learning rate scheduler with weight decay)
- Visualized plots and graphs of training/testing error rates and generated a confusion matrix visualization to analyze accuracy

ECommerce Full Stack Application | Java, SQL, JavaScript, JDBC, HTML, CSS, Database Design, Eclipse, MySQL

Apr. 2024

- Developed an original ecommerce platform using Java and MySQL for users to login, buy, sell, and search for products
- Designed a database and integrated it with software, using SQL to query and insert products for user functionalities.

Photo Gallery Applications | JavaFX, FXML, AndroidXML, SceneBuilder, Git

Apr. 2024

- Built two full stack applications, compatible with any Java IDE and Android devices, that allow users to view photos, and add dates, captions, tags, as well as save and copy to other albums.
- Designed Java controllers and Android Activities to handle GUI structures for front-end experience
- Implements Serializable interface to save/load photos, albums, and user data from previous sessions